NASIM ESHAGHIAN

AI and Machine Learning Engineer | Workflow Automation

n.eshaghian.sut@gmail.com

PROFILE

AI/ML engineer with experience developing industrial decision-support models, Persian speech recognition, retrieval-augmented assistants, and CRM automation. Combines Python and SQL development with temporal model evaluation, API integration, and technical documentation. M.Sc. graduate with a peer-reviewed publication in misinformation detection.

TECHNICAL SKILLS

Machine learning and data: Python, pandas, NumPy, scikit-learn, Matplotlib, classification, regression, feature engineering, time-series validation, baseline benchmarking.

AI and automation: Whisper, retrieval-augmented generation, embeddings, vector search, Supabase, OpenAI APIs, n8n, Telegram bots, REST APIs, webhooks, OpenAPI/Swagger.

Software and databases: PostgreSQL, SQL Server, SQL, Git/GitHub, Streamlit, web crawling, MATLAB, Weka; hands-on Redis/Memurai configuration.

PROFESSIONAL EXPERIENCE

Applied AI/ML R&D | Pardazesh Seir Afagh / Aqua Intelligent Technology

2025–Present | Water-treatment decision support

- Led model development and experimental evaluation for turbidity forecasting, coagulant decision support, and chlorine-demand/free-residual prediction using real plant data from Split Lake, Canada.
- Built Python data-preparation workflows to align irregular sensor readings and manual observations, define next-day targets, and investigate missing data, rare events, and timestamp inconsistencies.
- Developed a turbidity spike classifier achieving 70% precision and 63.6% recall on a 38-observation validation set; a separate spike-magnitude experiment reduced MAE from 6.76 to 4.26 versus persistence.
- Developed and compared coagulant models; investigated limited dose variation and historical support to identify cases requiring operator review.
- Evaluated chlorine models chronologically against persistence, documented generalization failures, and retained baseline predictions when candidates did not improve performance.
- Developed a Streamlit advisory dashboard, model packaging and registry configuration; prepared launch scripts and checked application startup and model inference.
- Translated process requirements into configurable site settings and operator-review logic; presented findings to domain and software leads and prepared technical and audit documentation.

AI and Workflow Automation | Hermes

May 2025–Present | CRM and exhibition-management workflows

- Built CRM and operational workflows with n8n and PostgreSQL, integrating Telegram bots, APIs, and company/exhibition registration.
- Implemented linked and bulk registration database functions; built a dropdown-options API and automated OpenAPI/Swagger generation from active workflows.
- Developed import workflows and designed Excel validation, Persian-column mapping, staging, and confirmation; debugged database schema, upload, and webhook issues.
- Specified Persian RTL CRM interfaces and role/group access requirements; developed SEO-support workflows and configured public access to self-hosted n8n.

ADDITIONAL AI PROJECTS

Persian Speech Recognition | Pardazesh Seir Afagh

- Fine-tuned Whisper on a local GPU and investigated Khorasani-accent adaptation, including attempted collection of accent-specific training data.
- Compared local recognition quality, infrastructure requirements, and cost with hosted speech APIs; supported the subsequent move to the OpenAI API.

Pharmaceutical Information Assistant | Pardazesh Seir Afagh

- Built a RAG prototype by crawling official Iranian medication information, indexing content for semantic retrieval, and connecting retrieved information to conversational answers.
- Worked with drug information and interaction content; tested an avatar interface and assessed licensing and operating costs.

EARLIER EXPERIENCE

Web Administrator | Raham Entrepreneur Consultants Corporation

December 2019–April 2023

- Designed, maintained, and troubleshot websites; supported website performance and online visibility.

Laboratory Assistant | Shahrood University of Technology

2013–2014

- Supported practical instruction in Weka data mining, preprocessing, clustering, classification, feature selection, databases, operating systems, and computer networks.

IT Center Laboratory Assistant | Sharif University of Technology

2012

- Supported Windows and network administration, Active Directory and DNS configuration, and systems maintenance.

EDUCATION

M.Sc. in Information Security | Sadjad University of Technology

2016–2019 | GPA 18.12/20 | Thesis: Fake News Detection on Twitter

B.Sc. in Information Technology Engineering | Shahrood University of Technology

2010–2014 | GPA 16.16/20

Capstone: Brain-tumour boundary detection using skeletonization of CT images | Grade 19.5/20

RESEARCH AND SELECTED TECHNICAL PROJECTS

- Misinformation detection: investigated Twitter social and user-profile features with Ordered Weighted Averaging fusion; research led to a peer-reviewed journal publication.
- Additional projects: car-price prediction using Python (2022); three-layer neural-network image classification using Python (2023); MATLAB fake-news classification and weighted fusion (2020).

Publication: Salkhordeh Haghighi, M., & Eshaghian, N. (2020). Fake News Detection Based on Social Features by Ordered Weighted Averaging Fusion. IJICT Research, 12(4), 46–59. DOI: 10.52547/ijict.12.4.46.

LANGUAGES

Persian: Native | English: IELTS Academic 7.0, January 2023 (historical score)